St. Philomena's College (Autonomous), Mysore

PG Department of Studies and Research in Chemistry

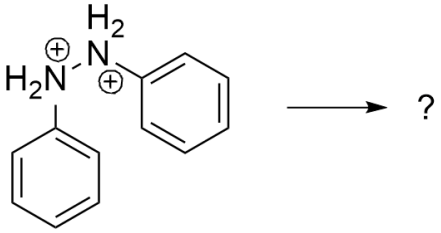
Question Bank (Revised Curriculum 2018 onwards)

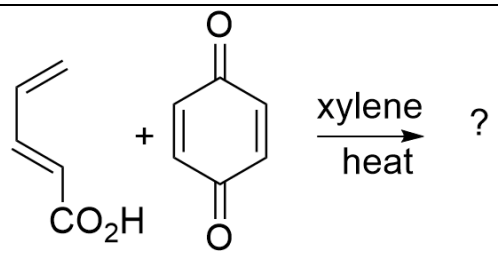
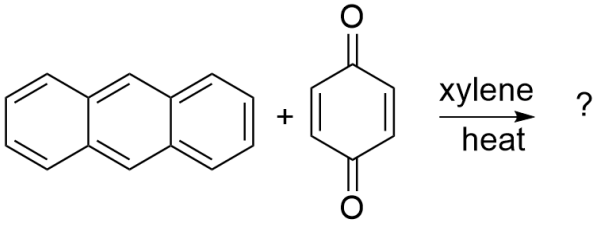
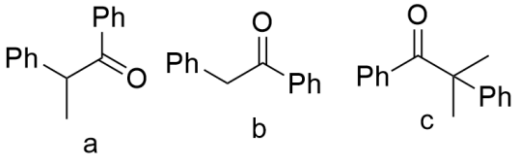
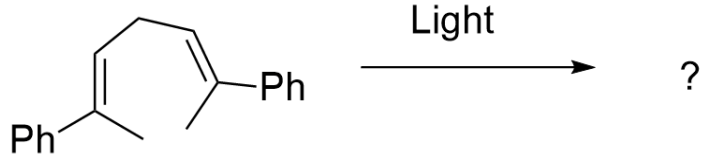
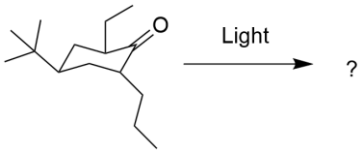
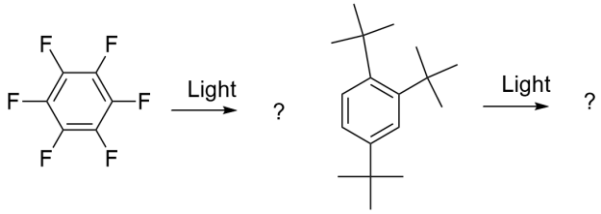
Second Year- Fourth Semester (2018 -20 Batch)

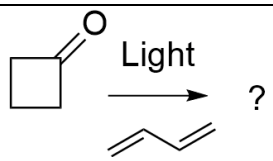
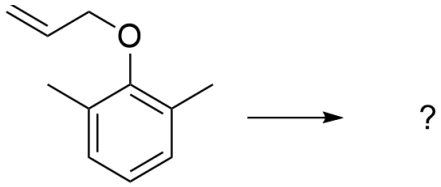
Course Title (Paper Title):

Photochemistry, Pericyclic Reactions and Organometallic Chemistry(SC) QP Code:55307

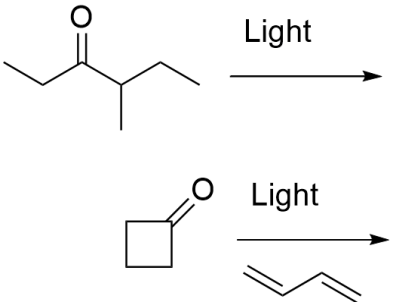
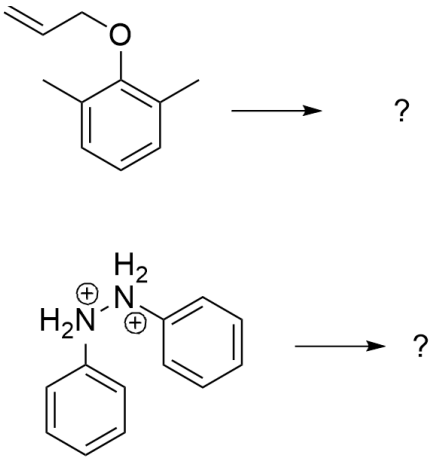
Unit	SI. No	Questions	Marks
1	1.	Define Sensitizer with suitable examples	2
1	2.	Give examples for [3,3] sigmatropic reactions.	2
1	3.	State woodword- Hoffman rule for Electrocyclic Reaction.	2
1	4.	Write briefly about Photochemistry of conjugated dienes.	2
1	5.	What is cope rearrangement? Give an example.	2
1	6.	Define Sigmatropic reactions with examples.	2
1	7.	How nitrile imines generated?	2
1	8.	what is photooxidation and photoreduction?	2
1	9.	Distinguish Diel's-Alder reaction from hetero Diel's- Alder reation	2
1	10.	Write an example for Paterno-Buchi reaction.	2
1	11.	What is Di-Pi Methane Rearrangement? Give an example.	2
1	12.	What are Pericyclic reactions? How are they classified?	2
1	13.	Define Sesitizers and Quenchers.	2
1	14.	Illustrate Aza-cope rearrangement with an example	2
1	15.	Mark HOMO and LUMO in molecular orbital diagram for hexatriene	2
1	16.	What happens when benzene is irradiated?	2
1	17.	What are pericyclic reactions? How they are classified?	2
1	18.	Distinguish Singlet and Triplet state.	2
1	19.	Define quencher and sensitizer with example.	2

1	20.	State Woodward-Hoffmann rules for 4n electrocyclic reactions.	2
1	21.	What is Di-Pi rearrangement? Give an example.	2
1	22.	How the different isomeric structure of benzene is formed, Name all the structures.	2
1	23.	State Woodward-Hoffmann rules for (4n+2) electrocyclic reactions.	2
1	24.	Draw HOMO and LUMO orbitals of 1,3-butadiene.	2
1	25.	What are dienophiles? Give example.	2
1	26.	Differenciate dienes from 1,3-dipoles.	2
1	27.	Define Sensitizer with suitable examples	2
1	28.	Give examples for [3,3] sigmatropic reactions.	2
1	29.	State woodword- Hoffman rule for Electrocyclic Reaction.	2
1	30.	Write briefly about Photochemistry of conjugated dienes.	2
1	31.	What is cope rearrangement? Give an example.	2
1	32.	Define Sigmatropic reactions with examples.	2
1	33.	How nitrile imines generated?	2
1	34.	Write the product for the following reaction. 	2
1	35.	Predict the product and name the reaction. 	2
1	36.	Predict the product and name the reaction.	2

			
1	37.	Predict the product and name the reaction. 	2
1	38.	Arrange the given compounds in increasing order of their reactivity when irradiated with light: 	2
1	39.	Deduce the product and name the reaction. 	2
1	40.	Envisage the product: 	2
1	41.	Envisage the products: 	2
1	42.	Predict the product and name the reaction.	2

			
1	43.	Predict the product and name the reaction. 	2
2	44.	Which product is obtained when organoaluminium reacts with allyl vinyl ether? Write the reaction.	2
2	45.	Point out the applications of Grignard reagent.	2
2	46.	Outline the preparation of organosilicon compounds	2
2	47.	Write any two synthetic applications of sulphur ylides	2
2	48.	What is Barton reaction? Give an example.	2
2	49.	What happens when an organo borate reacts with aldehyde and ketone?	2
2	50.	Write Peterson reaction.	2
2	51.	What are ylides? Give an example.	2
2	52.	Give an example for Barton decarboxylation reaction.	2
2	53.	Give examples for synthesis of organo silicon compounds.	2
2	54.	Give the preparation of organo sulphur compound.	2
2	55.	Write an example for Heck reaction.	2
2	56.	Give the synthesis of Organoaluminium compound.	2
2	57.	How dithianes are synthesized?	2
2	58.	Write any two applications of Organo aluminium compound.	2
2	59.	What is Heck reaction? Give an example.	2
2	60.	Mention the applications of sulphur ylides.	2
2	61.	Write the applications of hydroboration.	2

2	62.	Define Barton McCombie reaction and write the reaction involved in it.	2
2	63.	Give any two synthetic applications of Organolithium.	2
2	64.	Write any two preparations of organolithium	2
2	65.	Give any two synthetic applications of Organomagnesium.	2
2	66.	Outline the preparations of organolithium	2
2	67.	Give the reaction of Organoaluminium with α,β-unsaturated ketones	2
2	68.	Give the carbonylation reaction of alkylboranes.	2
2	69.	Give any two synthetic applications of ylides.	2
2	70.	Outline the preparations of organosulphur.	2
2	71.	How are dithiane synthesized?	2
2	72.	Write any synthetic applications of Organosilicon compound.	2
2	73.	Give the application of organoaluminium in synthesis of natural products.	2
2	74.	Give the Keck stereoselective allylation of organotin.	2
2	75.	Give the synthetic applications of Organosulphur.	2
2	76.	Write any reaction of Organoboron compound.	2
2	77.	Outline the preparations of Organoboron compound.	2
2	78.	Give an example for Barton oxygenation reaction.	2
2	79.	What is hydroboration? Give its application.	2
2	80.	What are the precautions to be taken in the preparation of Grignard reagent.	2
2	81.	Give the alkylation reaction of organoboron compounds.	2
2	82.	Write any two synthetic applications of organotin compounds.	2
2	83.	Give the reaction of Organoaluminium with acid chlorides.	2
2	84.	Give the synthetic applications of organotin.	2

2	85.	Write the acylation reaction of Organoboron.	2
2	86.	Give the reaction of alkyl halides with Organomagnesium.	2
2	87.	What are organosilicons? Give an example	2
2	88.	Outline the synthetic applications of Organolithium compound.	2
1	89.	Write all posible products for: 	**4**
1	90.	Predict the product and name the reaction. 	**4**
1	91.	How Nitrile oxides are generated? Write three synthetic applications.	4
1	92.	Among cis and trans alkene which one is more stable under photochemical condition and why?	4

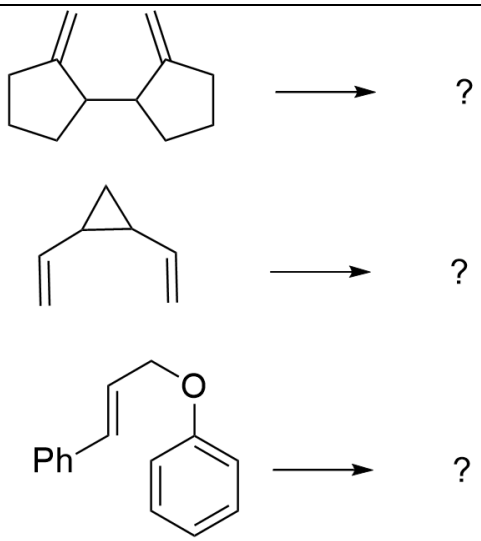
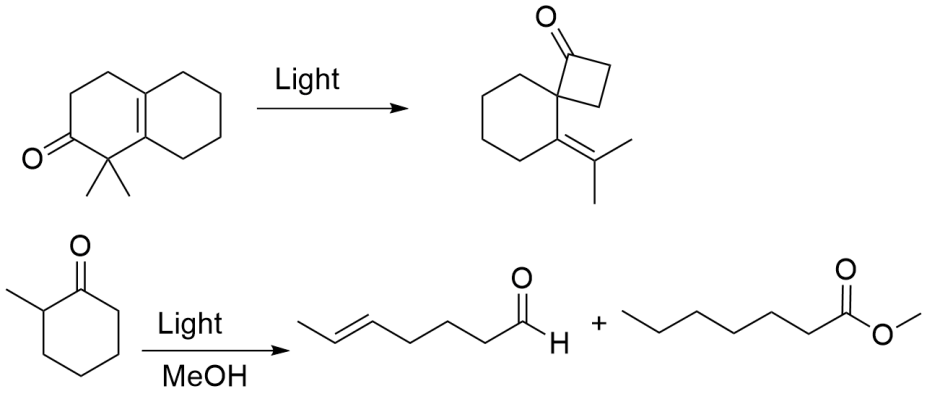
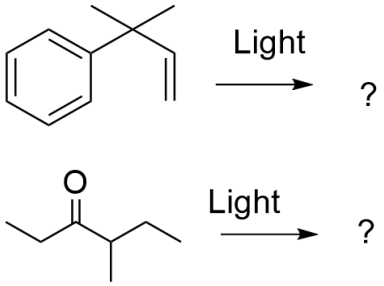
1	93.	Write a neat sketch of Jablonski diagram. Outline the mechanism of Paterno-Buchi reaction	4
1	94.	Outline intra and inter molecular 3+2 cycloaddition reaction and mention its application.	4
1	95.	With an example explain Photooxidation Reaction.	4
1	96.	Discuss Norrish type I with suitable examples.	4
1	97.	Discuss (5,5) sigmatropic rearrangement reactions.	4
1	98.	Discuss stereochemistry involved in Paterno-Buchi reactions.	4
1	99.	Illustrate Di-Pi methane rearrangement.	4
1	100.	Define Barton reaction and mention its application.	4
1	101.	What is cope rearrangement? Taking example ellucidate its mechanism	4
1	102.	How do you identify intermediates in photochemical reactions? Explain with an example	4
1	103.	Discuss Cis-trans isomerization of alkenes with suitable example.	4
1	104.	Compile the pathways possible for Norrish type 2 reaction by taking suitable examples	4
1	105.	Compile the pathways possible for Norrish type 1 reaction by taking suitable examples	4
1	106.	Illustrate the Photohemistry of Vision.	4
1	107.	Deduce woodward- Hoffmann rules for Electrocyclic Reactions	4
1	108.	State woodward- Hoffmann rules for (4+2) cycloaddition Reactions	4
1	109.	Cisoid geometry of Dienes are suitable for cycloaddition reaction. Justify	4
1	110.	Eloborate on stereochemistry of Diel's Alder Reaction.	4
1	111.	Illustrate synthetic application of Diel's Alder reaction.	4
1	112.	How Nitrile imines are generated? Write three synthetic applications.	4
1	113.	How Nitrile ylides are generated? Write three synthetic applications.	4

1	114.	Elobrate on [3,3] sigmatropic rearrangement with examples.	4
1	115.	State and prove woodward- Hoffmann rules for (2+2) cycloaddition Reactions using FMO diagram	4
1	116.	Dipict synthesis and applications of organoboron compounds.	4
1	117.	Dicuss the preparation and synthetic application of Grignard reagent.	4
1	118.	Write a note on hydroboration and its importance in organic synthesis.	4
1	119.	Give the preparation and synthetic application of organo lithium compounds.	4
1	120.	Briefly explain Suzuki coupling reaction.	4
1	121.	Illustrate with example the acylation reaction with organoboron compound.	4
1	122.	Write a note on Stille-Kelley coupling reaction.	4
1	123.	Outline the synthetic applications of organoaluminium compounds.	4
1	124.	Explain oxidation, protonolysis and carbonylation reaction of organo boranes.	4
1	125.	Discuss the prepation and synthetic applications of Grignard reagent.	4
1	126.	What is Heck reaction? Discuss its mechanism.	4
1	127.	Write a note on site selective and stereoselctive additions of nucleophiles mediated by organoaluminum reagents.	4
1	128.	Give the reaction of organomagnesium with aldehydes and ketones.	4
1	129.	How do you prepare aldehyde and ketone from organoboranes.	4
1	130.	Write a note on Baarton deoxygenation reaction.	4
1	131.	Write hydroboration reactions.	4
1	132.	Give synthetic applications of dithiane.	4
1	133.	Give any two applications of organoaluminium compound.	4
1	134.	Write any two preparation and synthetic applications of organo silicon compounds.	4

1	135.	Illustrate Heck reaction.	4
1	136.	Describe the Keck stereoselective allylation of organotin.	4
1	137.	Give any two synthetic applications of Organosulphur	4
1	138.	Write any two synthetic applications of Organotin	4
1	139.	Explain Barton decarboxylation reaction.	4
1	140.	Illustrate Stille coupling reaction of organotin reagents.	4
1	141.	Mention any four synthetic applications of Organomagnesium	4
1	142.	Give the preparations and synthetic applications of Organosulphur compounds	4
1	143.	Give preparation and synthetic applications of organosilicon compounds	4
1	144.	Write a note on Photochemistry of aromatic compounds.	5
1	145.	Explain 4+2 cyclo addition reactions.	5
1	146.	Explain Jablonski diagram with all the terms involved in it.	5
1	147.	what are pericyclic reactions? Mention the types with examples.	5
1	148.	State and prove woodward- Hoffmann rules for (2+2) cycloaddition Reactions using Correlation diagram	5
1	149.	Using FMO diagram show that Diels Alder reaction is thermally allowed reaction.	5
1	150.	Justify using correlation diagram that 4+2 cycloaddition is thermally allowed.	5
1	151.	Discuss the mechanism of Paterno Buchi Reaction	5
1	152.	State and prove Wood Ward Hofmann rule for (4n+2) electro cyclization using Correlation diagram.	5
1	153.	State and prove Wood Ward Hofmann rule for (4n+2) electro cyclization using FMO diagram.	5

1	154.	State and prove Wood Ward Hofmann rule for (4n) electro cyclization using FMO diagram	5
1	155.	Discuss the mechanism of Barton Reaction taking suitable example	5
1	156.	State and prove Wood Ward Hofmann rule for (4n) electro cyclization using Correlation diagram	5
1	157.	Explain by FMO method that s a 2π(suprafacial)+2π(antarafacial) cycloaddition is thermally allowed and photochemically forbidden?	5
2	158.	Explain Heck reaction with example.	5
2	159.	Write a note on Suzuki coupling.	5
2	160.	Give the oxidation and carbonylation reaction of organoboron compounds.	5
2	161.	Write the reaction of organoboron compounds with aldehydes and ketones	5
2	162.	Outline the preparations and synthetic applications of Organosulphur compounds	5
2	163.	Give the preparations and reactions of Organomagnesium compounds	5
2	164.	Write the preparations and synthetic applications of Organoaluminium compounds	5
2	165.	Discuss the preparations and reactions of Organoboron compounds	5
2	166.	Illustrate the preparations and reactions of Organosilicon compounds	5
2	167.	Outline the preparations and reactions of Organolithium compounds	5
2	168.	Write a note on hydroboration. Give its applications.	5
2	169.	Outline the preparations and synthetic applications of Organomagnesium compounds	5
2	170.	Give the preparations and reactions of Organosulphur compounds	5
2	171.	Write the synthesis and applications of Organosulphur compounds	5
2	172.	Outline preparation and synthetic applications of organosilicon compounds	5

1	173.	Discuss the mechanism invlved in Norrish type I and II reactions taking suitable examples.	6
1	174.	Deduce Woodward-Hoffmann rules for 4n and 4n+2 systems by FMO method.	6
1	175.	Using M.O. symmetry correction diagram prove that photochemically disrotatory ring closure of hexatriene to cyclohexadiene is a forbidden or allowed process.	6
1	176.	During electrocyclic reaction, which orbital symmetry is involved in Con rotator motion? Illustrate.	6
1	177.	Explain (3,3) and (5,5) sigmatropic rearrangement Reactions.	6
1	178.	What are cycloaddition reactions? How they are classified? Give atleast one example for each case.	6
1	179.	Deduce Woodward- Hoffmann rule for electrocyclic reaction 4n+2 electrons system under thermal and photochemical codition.	6
1	180.	Deduce Woodward- Hoffmann rule for electrocyclic reaction 4n electrons system under thermal and photochemical condition.	6
1	181.	State and prove Wood Ward Hofmann rule for 4n electro cyclization using FMO and Correlation diagram.	6
1	182.	Deduce Woodward-Hoffmann Rules for (2+2) cycloaddition reaction by FMO and Orbital correlation diagram.	6
1	183.	Deduce Woodward-Hoffmann Rules for (4+2) cycloaddition reaction by FMO and Orbital correlation diagram	6
1	184.	How Nitrile imines and Nitrile ylides are generated? Write two synthetic applications of each.	6
1	185.	Deduce the products with mechanism	6

			
1	186.	Provide a detailed mechanism for the following transformation: 	6
1	187.	Provide a detailed mechanism for the following transformation: 	6
2	188.	Explain briefly Suzuki coupling reaction.	6
2	189.	Write a note on Barton deoxygenaation reaction.	6

2	190.	Discuss Stille coupling reaction.	6
2	191.	Write a note on Sulphur Ylides and dithiane	6
2	192.	Discuss the preparation and synthetic applications of organolithium compounds.	6
2	193.	How do you synthesize the primary, secondary and territiary alcohols starting from Grignard reagent.	6
2	194.	Illustrate the preparation and reactions of organoaluminium compounds.	6
2	195.	Outline the synthesis of organomagnesium compounds. Mention its applications.	6
2	196.	Outline the synthesis of organoboron compounds. Mention its applications.	6
2	197.	Give the synthesis of organoboron compounds. Mention its applications.	6
2	198.	Discuss the preparation and synthetic applications of organo sulphur compounds.	6
2	199.	Illustrate the preparation and synthetic applications of organo silicon compounds.	6
2	200.	Write a note on Heck reaction.	6
2	201.	Outline the synthesis of primary, secondary and teritiary alcohols from organomagnesium compounds.	6
2	202.	What precautions to be taken in the preparation of Grignard reagent? Give any four synthetic applications.	6
2	203.	Depict synthesis and applications of organosilicon compounds	6

Question Paper Pattern- Model Question Paper

Q.P Code: 50443

St. Philomena's College (Autonomous) Mysore
IV Semester M.Sc. Final Examination : May 2019
Subject: CHEMISTRY
Title: Photo Chemistry, Pericyclic Reactions and Organometallic Chemistry (SC)

Time: 3 Hours **Max Marks: 70**

Instruction to the Candidates: ***Question paper has TWO parts answer both the parts. Provide clear and concise answer to the questions. Use chemical equations / drawings to support your answer where ever appropriate.***

PART –A

Answer the following Questions: **10×2=20**

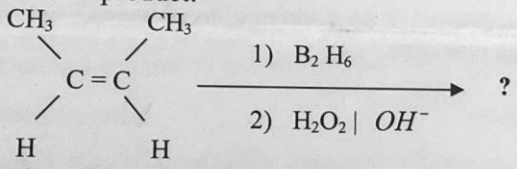
1. a. Define Diels Alder reaction.

   b. What are sigmatropic reactions? How are they classified?

   c. What are sensitizers? Give example.

   d. Give the synthesis of organo magnesium chloride.

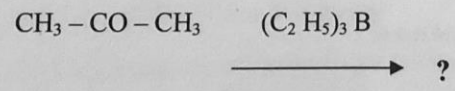
   e. Predict the product:



   f. Mention the applications of dithiane.

   g. Define Paterno – Buchi reaction with examples.

   h. Complete the following reaction:



   i. Give an example for 3 + 2, intermolecular cycloaddition.

   j. Give any two reactions of Butyl lithium.

PTO

PART –B

Answer any FIVE of the following: **5×10=50**

2. a. Explain Norvish type – I and II reactions.
   b. Describe the mechanism of Barton decarboxylation reaction. **5+5**
3. a. Explain Woodward – Hofmann rules for electrocyclic reactions taking examples.
   b. Discuss Jablonski diagram. **5+5**
4. a. Explain claisen and lope rearrangement.
   b. Discuss the mechanism of Suzuki coupling. **5+5**
5. a. Write a note on preparation and reactions of organo aluminium compounds.
   b. Explain 1, 3 – dipolar cycloaddition reactions involving nitrile oxide and nitrile-ylide. **5+5**
6. a. Discuss the mechanism of Stille – Kelley coupling reactions.
   b. Give the preparation and synthetic applications of organosulphur compounds. **5+5**
7. a. Write a note on: **3+3+4**
      i) Heck reaction
      ii) Peterson reaction
      iii) Di-Pi-rearrangement
8. a. Explain FMO theory of electrocylic reactions.
   b. Write notes on: **4+3+3**
      i) Photo reduction reaction
      ii) Reactions of organo lithium compounds

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